

HOMEWORK 7 SOLUTIONS

SECTION 3.3

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Suppose $\lambda=0$ is an eigenvalue of A . Then $0 = c_A(0) = \det(0 \cdot I - A) = \det(-A) = (-1)^n \det(A)$, so $\det(A) = 0$, implying that A is not invertible.

Conversely, if A is not invertible, then $\det(A) = 0$, so $c_A(0) = \det(0 \cdot I - A) = \det(-A) = (-1)^n \det(A) = 0$. Since $c_A(0) = 0$, we have that $\lambda=0$ is an eigenvalue of A .

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The characteristic polynomial of I is $c_I(x) = \det(xI - I) = \det((x-1)I) = (x-1)^n \det(I) = \boxed{(x-1)^n}$

The eigenvalues are the solutions to $c_I(x) = 0$.

$$c_I(x) = (x-1)^n = 0 \Leftrightarrow x-1 = 0 \Leftrightarrow x=1$$

So I has only one eigenvalue $\boxed{\lambda=1}$

The eigenvectors associated with $\lambda=1$ are the nonzero solutions $\vec{x}$ to $(1 \cdot I - I)\vec{x} = \vec{0}$, but $1 \cdot I - I = 0$, so every nonzero $\vec{x}$ in $\mathbb{R}^n$ is an eigenvector of I with eigenvalue 1.

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$$c_A(x) = |xI - A| = \begin{vmatrix} x-a & -b \\ -c & x-d \end{vmatrix} = (x-a)(x-d) - bc = x^2 - ax - dx + ad - bc = x^2 - (a+d)x + (ad-bc) = \boxed{x^2 - (\text{tr } A)x + \det(A)}$$

$\uparrow$ $\text{tr } A$ $\uparrow$ $\det A$

SECTION 5.1

1 a) $U = \{(1, s, t) \mid s, t \in \mathbb{R}\}$ is not a subspace because $\vec{0} = (0, 0, 0) \notin U$ (it is not of the form $(1, s, t)$).

c) $U = \{(u, v, w) \mid u, v, w \in \mathbb{R}, -u+3v+2w=0\}$ is a subspace. We check the three conditions.

• $\vec{0} = (0, 0, 0) \in U$ because $-0+3 \cdot 0+2 \cdot 0 = 0$.

• Let $\vec{x} = (u, v, w)$ and $\vec{y} = (a, b, c)$ be vectors in U . Is $\vec{x} + \vec{y} = (u+a, v+b, w+c) \in U$?

$$\left. \begin{array}{l} \vec{x} \in U \Leftrightarrow -u+3v+2w=0 \\ \vec{y} \in U \Leftrightarrow -a+3b+2c=0 \end{array} \right\} \Rightarrow -(u+a)+3(v+b)+2(w+c) = \underbrace{-u+3v+2w}_0 - \underbrace{a+3b+2c}_0 = 0 \Rightarrow \vec{x} + \vec{y} \in U.$$

• Let $\vec{x} = (u, v, w) \in U$ and $a \in \mathbb{R}$. Is $a\vec{x} = (au, av, aw) \in U$?

$$\vec{x} \in U \Leftrightarrow -u+3v+2w=0 \Rightarrow -au+3av+2aw = a \underbrace{(-u+3v+2w)}_0 = 0 \Rightarrow a\vec{x} \in U$$

ALTERNATIVE WAY: Since $-r+3s+2t = [-1 \ 3 \ 2] \begin{bmatrix} r \\ s \\ t \end{bmatrix}$, we have $U = \text{null}([-1 \ 3 \ 2])$, so it is a subspace of $\mathbb{R}^3$.

f) $U = \{(2r, -s^2, t) \mid r, s, t \in \mathbb{R}\}$ is not a subspace:

Take $\vec{x} = (2 \cdot 0, -1^2, 0) = (0, -1, 0) \in U$ and $a = -1$. Then $a\vec{x} = (0, 1, 0) \notin U$. Let's justify this:

If $(0, 1, 0) \in U$, then there exist $r, s, t \in \mathbb{R}$ with $(0, 1, 0) = (2r, -s^2, t)$, so (in particular) $-s^2 = 1 \Rightarrow s^2 = -1$. This equation has no (real) solutions, so $(0, 1, 0) \notin U$.

3a

We have to see if every $\vec{x} \in \mathbb{R}^4$ is a linear combination of $(1, 1, 1, 1)$, $(0, 1, 1, 1)$, $(0, 0, 1, 1)$ and $(0, 0, 0, 1)$.

Let $\vec{x} = (a, b, c, d) \in \mathbb{R}^4$ be any vector in $\mathbb{R}^4$. We need to know if there are $t_1, t_2, t_3, t_4 \in \mathbb{R}$ with

$$\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = t_1 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + t_2 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} + t_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} + t_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \end{bmatrix}.$$

The above system is consistent! So $\vec{x}$ is a linear combination of the above vectors. We conclude:

$$\{(1, 1, 1, 1), (0, 1, 1, 1), (0, 0, 1, 1), (0, 0, 0, 1)\} \text{ span } \mathbb{R}^4.$$

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No, because $(2, 0, 3)$ is not in $U = \{(r, s, 0) \mid r, s \in \mathbb{R}\}$.

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Let $\vec{y} \in U = \text{span}\{\vec{x}_1, \dots, \vec{x}_n\}$. Then $\vec{y} = a_1\vec{x}_1 + a_2\vec{x}_2 + \dots + a_n\vec{x}_n$ for some $a_1, a_2, \dots, a_n \in \mathbb{R}$. We have

$$A\vec{y} = A(a_1\vec{x}_1 + a_2\vec{x}_2 + \dots + a_n\vec{x}_n) = a_1A\vec{x}_1 + a_2A\vec{x}_2 + \dots + a_nA\vec{x}_n = a_1\vec{0} + a_2\vec{0} + \dots + a_n\vec{0} = \vec{0}$$

as desired.

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a) $U \cap W$ is a subspace of $\mathbb{R}^n$:

• $\vec{0} \in U \cap W$ because $\vec{0} \in U$ and $\vec{0} \in W$.

• Let $\vec{x}, \vec{y} \in U \cap W$. Then $\vec{x}$ and $\vec{y}$ are both in U and W . Since U and W are subspaces, $\vec{x} + \vec{y} \in U$ and $\vec{x} + \vec{y} \in W$, so $\vec{x} + \vec{y} \in U \cap W$.

• Let $\vec{x} \in U \cap W$, $a \in \mathbb{R}$. Then $\vec{x} \in U$ and $\vec{x} \in W$. Since U and W are subspaces, $a\vec{x} \in U$ and $a\vec{x} \in W$, so $a\vec{x} \in U \cap W$.

b) $U+W$ is a subspace of $\mathbb{R}^n$:

• $\vec{0} \in U+W$ because $\vec{0} = \vec{0} + \vec{0}$, $\vec{0} \in U$ and $\vec{0} \in W$.

• Let $\vec{x}, \vec{y} \in U+W$. Is $\vec{x} + \vec{y} \in U+W$?

$$\left. \begin{array}{l} \vec{x} \in U+W \Rightarrow \vec{x} = \vec{x}_1 + \vec{x}_2 \text{ with } \vec{x}_1 \in U, \vec{x}_2 \in W \\ \vec{y} \in U+W \Rightarrow \vec{y} = \vec{y}_1 + \vec{y}_2 \text{ with } \vec{y}_1 \in U, \vec{y}_2 \in W \end{array} \right\} \rightarrow \vec{x} + \vec{y} = \underbrace{(\vec{x}_1 + \vec{y}_1)}_U + \underbrace{(\vec{x}_2 + \vec{y}_2)}_W \in U+W.$$

• Let $\vec{x} \in U+W$ and $a \in \mathbb{R}$. Is $a\vec{x} \in U+W$?

$$\vec{x} \in U+W \Rightarrow \vec{x} = \vec{x}_1 + \vec{x}_2 \text{ with } \vec{x}_1 \in U \text{ and } \vec{x}_2 \in W \Rightarrow a\vec{x} = \underbrace{a\vec{x}_1}_U + \underbrace{a\vec{x}_2}_W \in U+W$$

SECTION 5.2

1 b) Is $\{(1,1,1), (1,-1,1), (0,0,1)\}$ linearly independent?

Suppose $a, b, c \in \mathbb{R}$ are such that

$$a \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a+b \\ a-b \\ a+b+c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Then a, b, c are solutions to the system

$$\begin{cases} a+b = 0 \\ a-b = 0 \\ a+b+c = 0 \end{cases} \quad \left(\begin{array}{l} \text{This system only has the trivial} \\ \text{solution } a=b=c=0 \end{array} \right)$$

Then, $a=b=c=0$. We conclude:

$\{(1,1,1), (1,-1,1), (0,0,1)\}$ is independent.

d) Is $\{(1,1,0,0), (1,0,1,0), (0,0,1,1), (0,1,0,1)\}$ independent?

Suppose $a, b, c, d \in \mathbb{R}$ are such that

$$a \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} + d \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a+b \\ a+d \\ b+c \\ c+d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Then (a, b, c, d) is a solution to

$$\begin{cases} a+b = 0 \\ a+d = 0 \\ b+c = 0 \\ c+d = 0 \end{cases} \quad \left(\begin{array}{l} \text{This system has a basic solution} \\ \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \end{array} \right)$$

Since the above system has nonzero solutions (for instance $a=c=1, b=d=-1$), we conclude:

$\{(1,1,0,0), (1,0,1,0), (0,0,1,1), (0,1,0,1)\}$ is dependent.

2b

Is $\{\vec{x}+\vec{y}, \vec{y}+\vec{z}, \vec{z}+\vec{x}\}$ independent?

Suppose $a, b, c \in \mathbb{R}$ are constants with $a(\vec{x}+\vec{y}) + b(\vec{y}+\vec{z}) + c(\vec{z}+\vec{x}) = \vec{0}$. Then

$$(a+c)\vec{x} + (a+b)\vec{y} + (b+c)\vec{z} = \vec{0}$$

Since $\{\vec{x}, \vec{y}, \vec{z}\}$ is independent, we get

$$\begin{cases} a+c=0 \\ a+b=0 \\ b+c=0 \end{cases} \quad \left(\begin{array}{l} \text{This system only has the solution} \\ a=b=c=0 \end{array} \right)$$

Thus, $a=b=c=0$, so

$\{\vec{x}+\vec{y}, \vec{y}+\vec{z}, \vec{z}+\vec{x}\}$ is independent.

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$$b) \mathcal{U} = \left\{ \begin{bmatrix} a+b \\ a-b \\ b \\ a \end{bmatrix} \mid a, b \in \mathbb{R} \right\} = \left\{ a \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \mid a, b \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Therefore, $\{(1, 1, 0, 1), (1, -1, 1, 0)\}$ is a spanning set for $\mathcal{U}$. It is also independent because it is a set of two nonparallel vectors, so

$\{(1, 1, 0, 1), (1, -1, 1, 0)\}$ is a basis of $\mathcal{U}$ and $\dim \mathcal{U} = 2$.

e)

$$\mathcal{U} = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \mid a+b-c+d=0 \right\} = \left\{ \begin{bmatrix} -b+c-d \\ b \\ c \\ d \end{bmatrix} \mid b, c, d \in \mathbb{R} \right\} = \left\{ b \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \mid b, c, d \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

So $\{(-1, 1, 0, 0), (1, 0, 1, 0), (-1, 0, 0, 1)\}$ is a spanning set of $\mathcal{U}$. Is it also independent?

$$x \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} -x+y-z=0 \\ x=0 \\ y=0 \\ z=0 \end{cases} \Rightarrow x=y=z=0. \text{ Thus, the set is also independent. Therefore:}$$

$\{(-1, 1, 0, 0), (1, 0, 1, 0), (-1, 0, 0, 1)\}$ is a basis of $\mathcal{U}$ and $\dim \mathcal{U} = 3$.